

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [86, 78, 85, 77, 87, 73, 75, 72, 79, 65]

Tap amplitudes (pixels): [145.2067491544384, 159.20113064925135, 156.40012787718558, 150.60212481900777, 147.05441169852742, 158.3729774930054, 156.4736399525492, 149.5125412799876, 150.41608956491322, 154.01298646542764, 144.40221604947757]

GPT-5

MDS-UPDRS Item 3.4 – Finger Tapping Scoring Analysis

What the data shows

- Stated target: 10 taps; detected features show 11 amplitudes and 10 inter-tap periods (which is consistent: N amplitudes for N taps, N–1 periods).
- Periods (lower = faster): [86, 78, 85, 77, 87, 73, 75, 72, 79, 65] frames
 - Trend: overall slightly decreasing (86 → 65), indicating accelerating or at least no slowing. No clear “slight/mild/moderate slowing.”
 - Interruptions/hesitations: No obvious long outliers or spikes suggesting arrests; all periods are within a relatively tight band with a gentle downward drift.
- Amplitudes: [145.21, 159.20, 156.40, 150.60, 147.05, 158.37, 156.47, 149.51, 150.42, 154.01, 144.40] pixels
 - Early values: 145–159
 - Mid values: oscillate 147–158
 - Late values: 154.01 then 144.40
 - Pattern: variability but no consistent, progressive decrement beginning near start or mid-sequence. The final tap is lower (144.4) than some earlier taps, but late-sequence dips follow several rebounds (e.g., 158.37, 156.47, 154.01), so this looks like normal variability rather than a sustained decrement pattern.

Mapping to scoring criteria

- Interruptions/hesitations: Period series does not suggest 3–5 interruptions (score 2) nor >5 or a freeze (score 3). At most, there could be minor variability without clear pauses.
- Speed: No slowing; if anything, slight speeding up. Not compatible with “slight/mild/moderate slowing.”
- Amplitude decrement:
 - Score 3 requires decrement starting after the 1st tap: not present.
 - Score 2 requires decrement midway through the sequence: not sustained; amplitudes recover multiple times in mid-to-late taps.
 - Score 1 allows end-of-sequence decrement: there isn’t a clear monotonic end-of-run decline—there’s a final lower value, but immediately preceding taps are not trending down; rather they’re relatively high. This does not convincingly meet “decrements near the end.”

Conclusion

The tap train appears regular with consistent speed and amplitude, no clear hesitations, arrests, or sustained decrement.

Final score: 0
